# Exploratory Data Analysis: Investment and GDP Growth

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# 1 Introduction

This project examines the relationship between **Gross Capital Formation Rate** (investment) and **GDP per capita growth** across countries to determine if higher investment leads to stronger economic growth.

## 1.1 Research Question

Does a higher Gross Capital Formation Rate lead to higher GDP per capita growth?

## 1.2 Hypothesis

- $\mathbf{H}_0$: No significant relationship exists between investment and GDP growth.
- $\mathbf{H}_1$: Higher investment levels are associated with higher GDP growth.

# 2 Dataset Description

**Source:** World Bank (1990-2023)
**Variables:**

- **Investment Rate** (% of GDP)
- **GDP Growth** (annual % change)
- **Country** and **Year**

**Population:** Countries worldwide, covering 33 years of economic data.

# 3 Data Preparation

- Removed irrelevant columns (Country Code, Indicator Name, Indicator Code...).
- Removed NaN values.
- Reshaped data (Wide to Long format).
- Grouped Countries by Region.
- Categorized data by Income Group: Low, Lower middle, Higher middle and High income.
- Selected only the years from 1993 to 2023 for analysis.

# 4 Population

The population of interest in this analysis includes countries worldwide, from which economic data on Gross Capital Formation Rate (investment as a percentage of GDP) and GDP growth span from 1993 to 2023. The dataset includes over 190 countries, though filtering criteria (such as missing data on regions or income groups) may reduce the final sample size. The analysis Covers economic 33 years of data,

ensuring a long-term perspective on investment-growth relationships. Countries are grouped into six major regions: Latin America and Caribbean, East Asia and Pacific, Europe and Central Asia, Sub-Saharan Africa, South Asia, Middle East and North Africa. Countries are also categorized into four income groups based on World Bank classifications.

# 5 Variable Distributions

## 5.1 Average lagged Investment rate

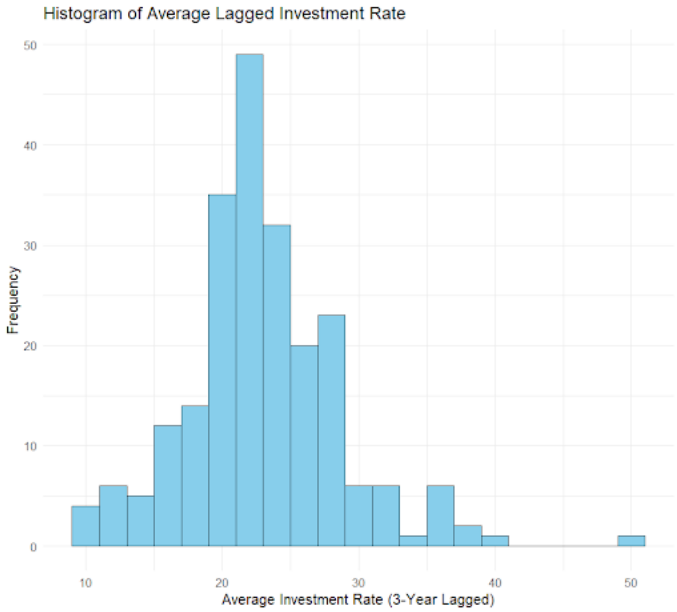



Figure 1: Figure 1: Histogram of Average Lagged Investment Rate (3-Year Lagged). The x-axis represents the Average Investment Rate (3-Year Lagged), while the y-axis represents the Frequency of occurrences in the dataset. The distribution appears to be right-skewed, with most observations concentrated around lower investment rates.

## 5.2 Density of Average GDP Growth

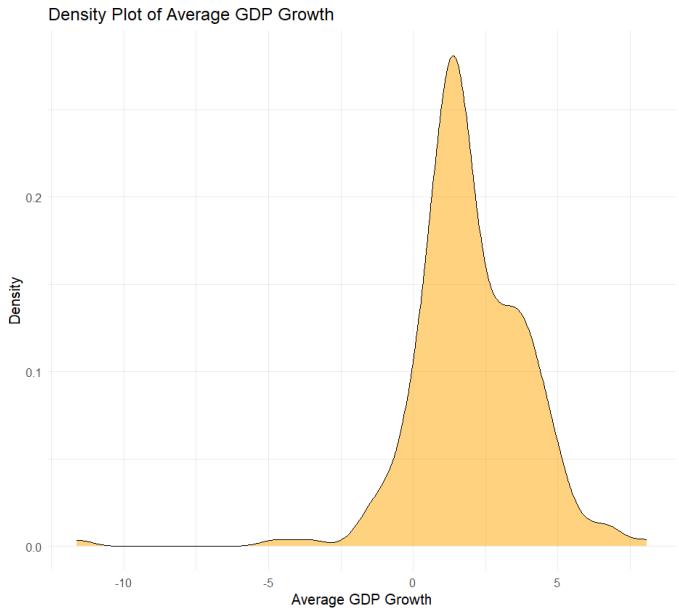



Figure 2: Density plot of average GDP growth rates across different regions. The x-axis represents the range of GDP growth rates, while the y-axis shows the density, indicating how common each growth rate is within the dataset. This visualization highlights the distribution and variability in economic performance.

## 5.3 Histogram Average GDP Growth

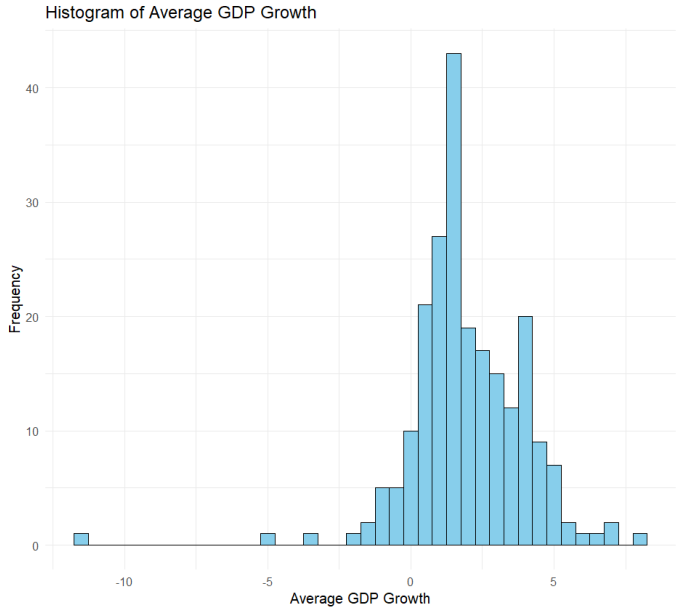



Figure 3: Histogram of average GDP growth rates across different regions. The x-axis represents the range of GDP growth rates, divided into intervals or bins, while the y-axis shows the frequency or count of occurrences within each bin. This visualization provides insight into the distribution and concentration of economic growth rates.

### 5.4 Average GDP Growth vs Lagged Investment rate (country-level Averages

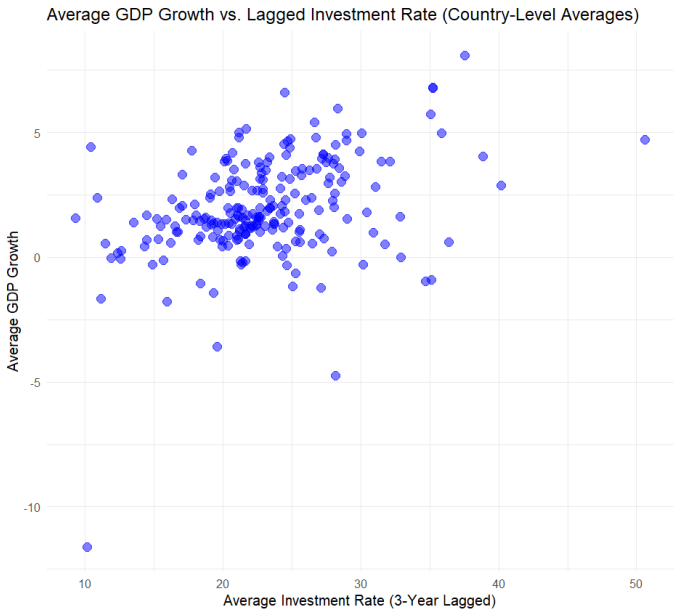



Figure 4: Scatter plot of average GDP growth versus lagged investment rate at the country level. The x-axis represents the average investment rate (lagged by three years), while the y-axis shows the average GDP growth rate. Each point corresponds to a country, illustrating the relationship between past investment rates and subsequent economic growth.

## 6 Group Analysis: Effective Investment Rate by Region

### 6.1 Sub-Saharan Africa

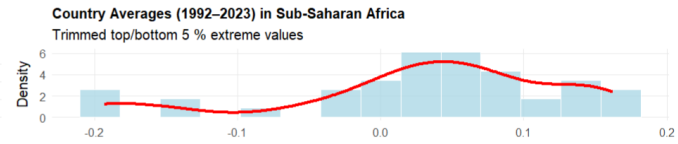



Figure 5: Distribution of effective investment rate in Sub-Saharan Africa from 1993 to 2023. The x-axis represents the effective investment rate, while the y-axis shows the frequency of density within each investment rate over 30 years.

The graph shows a slight right skew with moderate clustering  0.05 to 0.15. The distribution suggests moderate consistency in investment rates.

## 6.2 South Asia

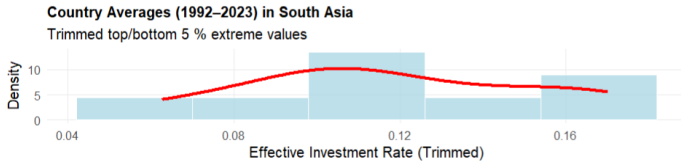



Figure 6: Distribution of effective investment rate in south Asia from 1993 to 2023. The x-axis represents the effective investment rate, while the y-axis shows the frequency of density within each investment rate over 30 years.

We can appreciate tight clustering between 0.8 and 0.14. This region has the highest density peal among the graphs, showing how the region is more uniform than the rest.

## 6.3 Middle East and North Africa

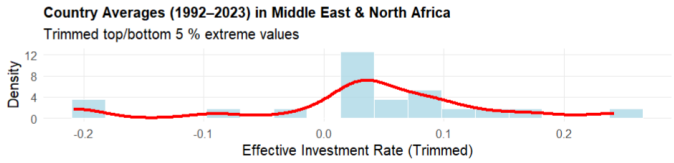



Figure 7: Distribution of effective investment rate in the Middle East and North Africa from 1993 to 2023. The x-axis represents the effective investment rate, while the y-axis shows the frequency of density within each investment rate over 30 years.

The graph is fairly symmetrical with a moderate peak. However, the investment rates mostly hover between 0 and 0.15.

## 6.4 Latin America and Caribbean

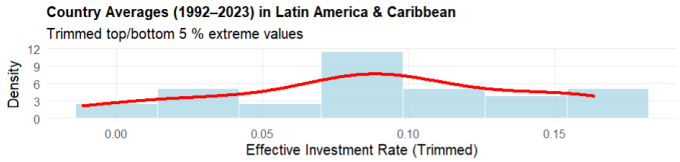



Figure 8: Distribution of effective investment rate in Latin America and Caribbean from 1993 to 2023. The x-axis represents the effective investment rate, while the y-axis shows the frequency of density within each investment rate over 30 years.

It is a clear right-skewed distribution, where most values are under 0.1. The high peak suggests clustering of countries around a low investment rate.

## 6.5 Europe and Central Asia

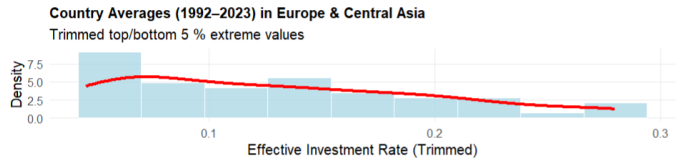



Figure 9: Distribution of effective investment rate in Europe and Central Asia from 1993 to 2023. The x-axis represents the effective investment rate, while the y-axis shows the frequency of density within each investment rate over 30 years.

In the case of Europe and Central Asia, we find a flatter distribution, hinting a wider spread of investment rates. This shows more variation across countries.

## 6.6 East Asia and pacific

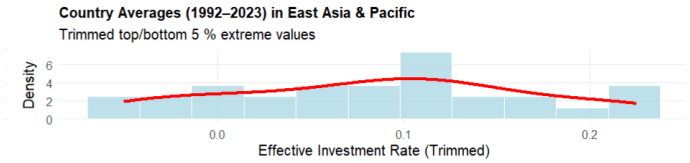



Figure 10: Distribution of effective investment rate in East Asia and Pacific from 1993 to 2023. The x-axis represents the effective investment rate, while the y-axis shows the frequency of density within each investment rate over 30 years.

Finally, the graph for this region shows a bell shaped, which makes it the closest to a normal distribution, slightly right skewed distribution.

# 7 Group Analysis: Effective Investment Rate by Income Group

## 7.1 Country-Level Effective Investment Rates by Income Group




Figure 11: Kernel Density Estimation (KDE) Effective investment rate (by income group) from 1993 to 2023.

This figure displays the distribution of the effective investment rate as a percentage of GDP across four income groups: Low income, Lower middle income, Upper middle income, and High income. The distributions indicate differences in investment behaviors across income groups. As income level rises, the distribution of effective investment rates becomes more symmetric, centered, and consistent. Low-income countries show dispersed, unpredictable outcomes, whereas high- and upper-middle-income groups exhibit tight, unimodal distributions, indicating greater uniformity and investment efficiency.

## 7.2 Comparison Across Income Groups and Regions




Figure 12: Effective Investment Rate by Income Group and by Region from 1993-2023.It represents the median, interquartile range, and variety of investment rates.

We can appreciate results in different income groups vary: higher-income groups tend to have more concentrated and stable investment rates, while low income groups show the greatest variability, the lowest median and even negative values.

In the case of the Invesment Rate by region, we can appreciate that South Asia and Europe & Central Asia stand out as regions with more efficient investment use. North America appears to invest with low efficiency, though consistently. Lastly, Sub-Saharan Africa and Middle East & North Africa display investment challenges, likely reflecting infrastructure or institutional weaknesses.

The data was trimmed to exclude extreme outliers to make patterns clearer.

# 8 Variable characteristics: Continuous vs. discrete

This study investigates whether a higher Gross Capital Formation Rate (investment rate) leads to higher GDP per capita growth. To analyze this, we use two key variables:

Investment Rate (GDP Percentage):

- Definition: The percentage of a country's GDP allocated to investments (Gross Capital Formation).
- Type: Continuous (measured as a percentage, can take any value within a range).
- Role in the study: Independent variable (predictor).
- Expected Relationship: Higher investment rates may lead to increased GDP growth.

GDP Growth (annual percental change)

- Definition: The annual percentage change in a country's GDP per capita.
- Type: Continuous (measured as a percentage, with decimal values).
- Role in the study: Dependent variable (outcome).
- Expected Relationship: If investment drives economic growth, higher investment rates should be associated with higher GDP growth.

Country:

- Definition: The nation under study.
- Type: Discrete (categorical, represents different countries).

- Role in the study: Grouping variable for analysis.

Year:

- Definition: The specific year for which data is recorded (1990-2023).
- Type: Discrete (integer values representing years).
- Role in the study: Used to track trends over time.

Region:

- Definition: The geographical region where a country belongs (e.g., Sub-Saharan Africa, East Asia and Pacific).
- Type: Discrete (categorical).
- Role in the study: Allows regional comparisons.

Income Group:

- Definition: World Bank classification of a country's income level (Low, Lower-Middle, Upper-Middle, High).
- Type: Discrete (categorical).
- Role in the study: Helps analyze investment-GDP growth patterns across income levels.

## 8.1 Justification for Variable Selection

In our analysis, we focus on two primary variables: GDP per capita growth and the investment rate as a percentage of GDP. We use GDP per capita growth rather than total GDP growth to isolate economic performance from demographic effects. This approach ensures that any observed growth is not simply due to population increases, but rather reflects improvements in productivity and income levels at the individual level.

The investment rate, measured as a share of GDP, serves as a proxy for the resources allocated toward future economic output. This includes capital formation, infrastructure, and technological development. By relating this measure to GDP per capita growth, we aim to capture the effectiveness of investment in promoting economic development.

We chose to explore lagged investment rates to account for the delayed effects of capital accumulation on economic performance. Investments, especially in infrastructure or technology, often take time to translate into measurable economic

growth. Lag structures allow us to assess whether prior investments have a statistically significant influence on subsequent growth.

We considered using the growth rate of investment, but concluded that the level of investment relative to GDP provides a more stable and interpretable measure of a country's commitment to capital formation. This distinction is particularly relevant when comparing countries at different stages of development.

In lower-income countries, investment is expected to primarily contribute to increases in the capital stock, such as machinery, roads, and basic infrastructure, leading to direct increases in output. In contrast, high-income countries typically allocate a larger share of investment toward innovation, human capital, and productivity-enhancing technologies. Thus, in more developed economies, investment aims to drive growth through efficiency gains rather than sheer expansion of capital.

This framework allows us to explore how investment, in both quantity and timing, correlates with economic performance across different income levels.

# 9 Statistical Measures

- Statistical measures for the Effective Investment rate

Mean, variance, and standard deviation were computed for The Effective Investment rate. Where the Effective Investment rate is: $EIR_t = \frac{GDP_Growth_t}{Investment_Rate_{t-3}}$.

This is the mean, variance and standard deviation for each country: Download df_country_avg.csv

| | Mean | Variance | Standard deviation |
|---|---|---|---|
| World | 0.0726 | 0.00452 | 0.0673 |

Table 1: World Summary statistics (1993-2023)

- Statistical measures for the Effective Investment rate by groups

| Region | Mean | Variance | Standard deviation | Nº of Countries |
|---|---|---|---|---|
| Latin America and Caribbean | 0.0967 | 0.0520 | 0.228 | 28 |
| East Asia and Pacific | 0.0932 | 0.0976 | 0.313 | 29 |
| Europe and Central Asia | 0.135 | 0.0964 | 0.310 | 51 |
| Sub-Saharan Africa | 0.0966 | 4.264 | 2.0648 | 41 |
| South Asia | 0.132 | 0.0187 | 0.137 | 8 |
| Middle East and North Africa | 0.112 | 2.072 | 1.439 | 20 |
| North America | 0.0631 | 0.0183 | 0.135 | 3 |

Table 2: Summary statistics by region (1993-2023)

| Income | Mean | Variance | Standard deviation | Nº of Countries |
|---|---|---|---|---|
| Low income | 0.131 | 8.275 | 2.877 | 22 |
| Lower middle income | 0.0901 | 0.0328 | 0.181 | 45 |
| Upper middle income | 0.081 | 0.194 | 0.439 | 50 |
| High income | 0.088 | 0.056 | 0.236 | 63 |

Table 3: Summary statistics by Income (1993-2023)

# 10 Theoretical probability distributions

The possible distributions we can see while plotting the density functions of the effective investment rate among countries could be log-normal distribution or gamma distribution which graphs are shown above.

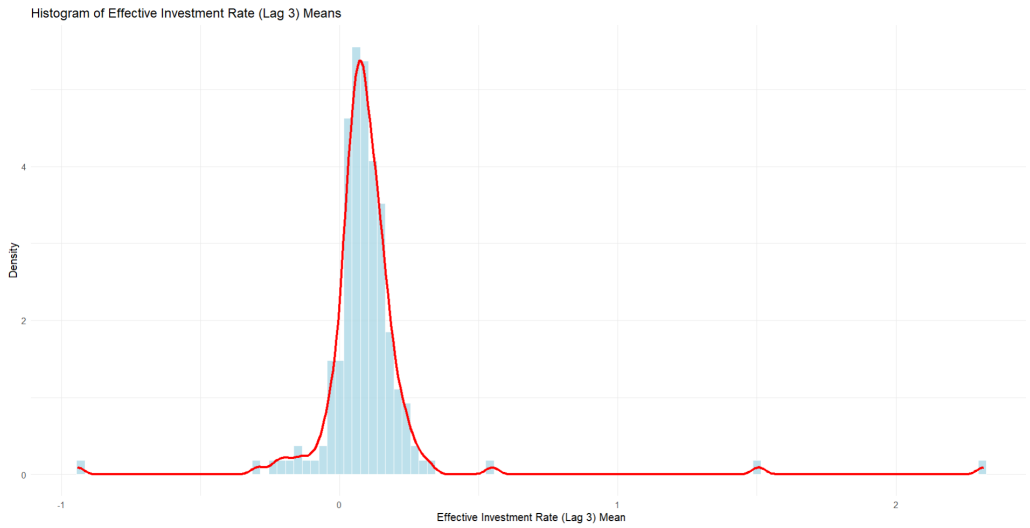



Figure 13: Effective Investment rate Histogram

Before testing our hypothesis about investment and GDP growth, we checked whether our variables are normally distributed using the Shapiro-Wilk test. The result ($p = 0.7998$) suggests the distribution does not significantly deviate from normality, so we proceed with other models that may assume it.

# 11 Hypothesis testing

## 11.1 Linear Regression

### 11.1.1 Hypothesis

- Null Hypothesis $H_0$: There is no significant linear relationship between the 3-year lagged investment rate and GDP per capita growth.
- Alternative Hypothesis $H_1$: There is a significant linear relationship between the 3-year lagged investment rate and GDP per capita growth.

### 11.1.2 Test and rationale

We performed a simple linear regression test where GDP per capita growth was the dependent variable, and the 3-year lagged investment rate was the independent

variable. This method was chosen to evaluate whether higher past investment rates predict GDP growth, assuming a linear relationship.

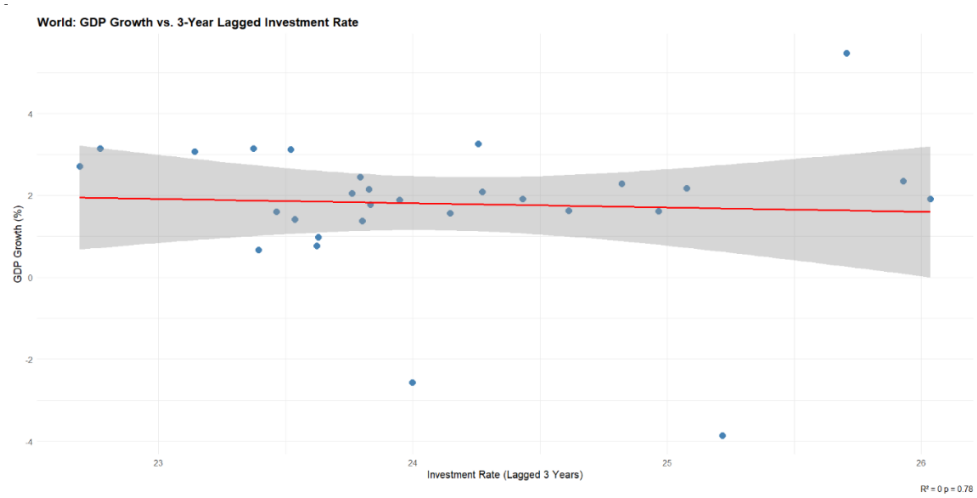



Figure 14: GDP Growth vs 3-Year Lagged Investment Rate

### 11.1.3 Results

- Slope: slightly negative, suggesting a very weak inverse relationship
- R-squared = 0.0029. It indicates that only 0.29% of the variation in GDP per capita growth is explained by the investment rate, effectively suggesting no meaningful linear relationship between the variables.
- p-value = 0.78. This high p-value indicates that the slope of the regression line is not statistically significant at any conventional level.
- Confidence Interval: the gray band around the regression line is wide and flat, further suggesting high uncertainty and weak predictive power.

### 11.1.4 Interpretation

The linear regression analysis suggests no statistically significant relationship between the lagged investment rate and GDP per capita growth. The model shows a nearly flat trend line, and the R-squared value of nearly 0.0 indicates that the investment rate does not explain changes in GDP growth in the data set.

From this data, we can fail to reject our Null Hypothesis. $\mathbf{H}_0$: No significant relationship exists between investment and GDP growth.

## 11.2 Multivariate Regression by income group

### 11.2.1 Hypothesis

- **Null Hypothesis ($H_0$):** The relationship between investment and GDP growth is the same across all income groups.
- **Alternative Hypothesis ($H_1$):** The effect of investment on GDP growth varies by income group.

### 11.2.2 Test and Rationale

We conducted a multivariate linear regression to evaluate whether the impact of investment on GDP growth differs by income level. The model includes both continuous and categorical predictors, as well as interaction terms.

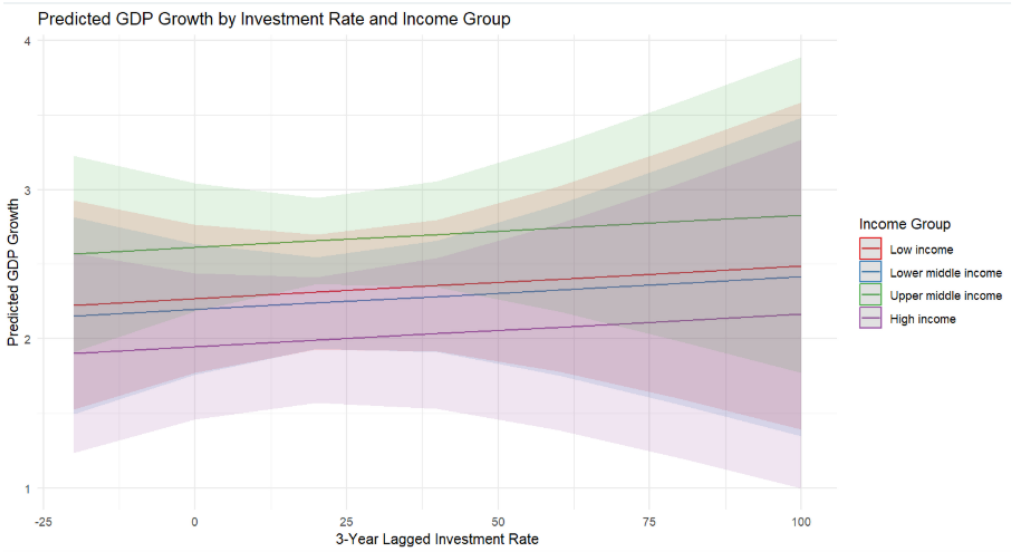



Figure 15: GDP Growth vs 3-Year Lagged Investment Rate in all Income groups

- **Dependent variable:** Annual GDP per capita growth (continuous)
- **Key independent variables:**
  - Investment_Rate_Lag3: Investment as a percentage of GDP, lagged by 3 years
  - IncomeGroup: Categorical variable (Low, Lower-Middle, Upper-Middle, High)
  - Interaction term: Investment_Rate_Lag3 × IncomeGroup
- **Controls:** Year and region fixed effects to account for time and geographic differences

This model tests for heterogeneous investment returns by allowing the slope of the investment-growth relationship to vary across income groups. It also incorporates lagged effects to reflect the delayed impact of investment.

### 11.2.3 Results

- **High-income countries:** A 1% increase in lagged investment corresponds to a ∼0.15% increase in GDP per capita growth.
- **Low-income countries:** A 1% investment increase yields only ∼0.05% growth, and can even produce negative growth when investment levels fall below 15–20% of GDP.
- **Middle-income countries:** Display intermediate effects, suggesting a transitional efficiency of capital.

### 11.2.4 Interpretation

The regression analysis supports the hypothesis that investment effectiveness differs across income groups.

- High-income countries show the strongest and most consistent returns to investment, likely due to more efficient institutions, infrastructure, and innovation systems.
- Low-income countries exhibit weak or even negative returns at lower investment levels, implying that investment alone is insufficient. Structural reforms—such as improving governance and human capital—may be necessary to enhance returns.
- The observed investment threshold (≈20% of GDP) in low-income countries suggests that small-scale efforts may be inadequate for growth.

These findings indicate that policy efforts should focus not just on increasing investment, but on improving its quality and institutional context, particularly in developing economies.

## 11.3 Multivariate Linear Regression by regions

### 11.3.1 Hypothesis

- **Null Hypothesis ($\mathbf{H}_0$):** The marginal effect of investment on GDP growth is the same across all regions.
- **Alternative Hypothesis ($\mathbf{H}_1$):** The effect of investment on GDP growth varies significantly across regions.

### 11.3.2 Test and Rationale

We estimated a multivariate linear regression model to test whether the effect of investment on GDP growth differs by region. The model incorporates both regional dummy variables and interaction terms with the investment variable.

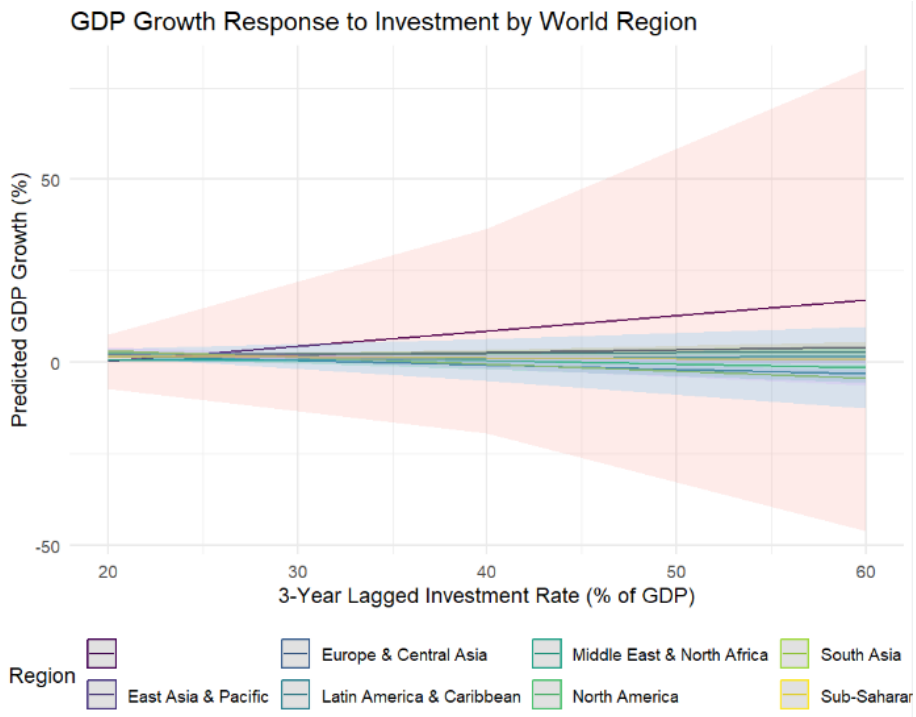



Figure 16: GDP Growth vs 3-Year Lagged Investment Rate in all Regions

- Dependent variable: Annual GDP per capita growth (continuous)
- **Key independent variables:**
  - Investment Rate Lag3 (continuous): Investment as a percentage of GDP, lagged by 3 years
  - Region (categorical): Geographic classification (e.g., Sub-Saharan Africa, East Asia & Pacific)
  - Interaction terms: Investment_Rate_Lag3 × Region
- **Controls:** Year fixed effects to account for global shocks or trends; optionally controlled for income group

The model allows for regional heterogeneity in investment returns and captures delayed impacts of capital formation through lagging the investment variable.

### 11.3.3 Results

- **East Asia & Pacific:** Highest marginal return to investment (0.12–0.15% GDP growth per 1% investment), indicating strong capital efficiency, likely due to export-led growth and sound institutions.

- **South Asia:** Also shows a consistently positive investment-growth relationship, supported by favorable demographics and rising technological integration.
- **Latin America & the Caribbean:** Moderate but volatile returns, potentially due to macroeconomic instability and commodity dependency.
- **Sub-Saharan Africa:** Lowest and occasionally negative marginal returns, suggesting that capital alone is insufficient without structural reforms.
- **North America and Europe & Central Asia:** High intercepts indicate strong baseline growth even with low investment levels, reflecting mature economies with productive capital stocks.

#### 11.3.4 Interpretation

The regression confirms that the effectiveness of investment varies by region, reinforcing the importance of institutional and policy contexts:

- **East Asia & Pacific:** Exemplifies high-ROI environments where sound governance, infrastructure, and export orientation amplify capital efficiency.
- **Sub-Saharan Africa:** Faces weak or negative investment returns due to underlying structural challenges like political instability, infrastructure gaps, and limited human capital.
- **Latin America:** Indicates difficulty in leveraging investment amidst policy volatility and economic dependence on primary exports.
- **North America and Europe:** Demonstrate that advanced economies can sustain growth through productivity and innovation, even with modest capital accumulation.

### 11.4 Correlation

#### 11.4.1 Hypothesis

- Null Hypothesis $H_0$: There is no correlation between the 3-year lagged investment rate and GDP per capita growth.
- Alternative Hypothesis $H_1$: Ther is a significant correlation betweeen the 3-year lagged investment rate and GDP per capita growth.

#### 11.4.2 Test and rationale

We used a Pearson Correlation test to meassure the strenght and direction of association between the two varibales GDP per caputa growth and the Investmenr

rate (3-year lag). This method assesses linear dependence and helps identify country specific patterns.

We conducted the test for 180 countries. Taking into account we used both the absolute value of the correlation coefficient and the p-value of the test, we reached the following conclusions. A strong correlation occurs when the p-value is $< 0.05$ and $|corr_coef| \geq 0.7$, a moderate correlation happens when the p-value is $< 0.05$ and $|corr_coef| \geq 0.4$, and when the p-value is $< 0.05$ and $|corr_coef| \geq 0.2$, the correlation is weak. Lastly, when the p-value is $< 0.1$, the correlation is very weak.

### 11.4.3 Results

- 5 countries showed strong correlation
- 27 countries showed a moderate correlation
- 15 countries showed weak correlation
- 14 countries showed very weak correlation
- 113 countries showed no correlation
- 6 countries had not enough observations to reach conclusions

Using the same criteria this were the results for the analysis of income groups.

### 11.4.4 Results for income groups

- For high income countries there is very weak correlation
- For upper middle income countries there is very weak correlation
- For lower middle income countries there is very weak correlation
- For low income countries there is weak correlation

### 11.4.5 Results for regions

- South Asia showed a weak correlation
- Europe and Central Asia showed a weak correlation
- East Asia and Pacific showed a very weak correlation
- Latin America showed a very weak correlation
- Middle East and North Africa showed no correlation
- North America showed a weak correlation
- Sub-Saharan Africa showed very weak correlation

### 11.4.6 Interpretation

The vast majority of countries show no meaningful correlation. Even where the correlation exists, it is weak or very weak. Therefore, we fail to reject the null hypothesis, supporting the regression result that investment and growth are not linearly associated.

## 11.5 Two-sample t-test

### 11.5.1 Hypothesis

- Null Hypothesis $H_0$: There is no significant difference in effective investment rate between high and low income countries.
- Alternative Hypothesis $H_1$: There is a significant difference in effective investment rate between these income groups.

### 11.5.2 Test and Rationale

We conducted a two-sample t-test comparing the means of high and low income countries. This test assesses whether the average investment rate differs significantly between groups.

### 11.5.3 Results

- Mean (High-income): 0.0882
- Mean (Low-income): 0.1306
- t-value: -0.35
- Degrees of freedom: 558.43
- p-value: 0.7285. This high p-value indicates that the observed difference in means is not statistically significant at any conventional level (e.g., 0.05).
- Confidence interval: The 95 % confidence interval for the difference in means ranges from -0.2820 to 0.1973, which includes zero, suggesting no clear evidence of a difference between the groups.

### 11.5.4 Interpretation

The two-sample t-test results suggest no statistically significant difference in the Effective Investment Rate between high-income and low-income countries. Although the mean investment rate is slightly higher in low-income countries

(0.1306) than in high-income countries (0.0882), the difference is small and not statistically meaningful.

Given the wide confidence interval and high p-value, we fail to reject the null hypothesis $H_0$: there is no significant difference in mean investment rates between the two income groups.

## 11.6 Chi-square test

### 11.6.1 Hypothesis

- Null Hypothesis $H_0$: Income group and investment rate category are independent.
- Alternative Hypothesis $H_1$: There is an association between income group and investment rate category.

### 11.6.2 Test and rationale

We used a chi-squared test for independence to determine whether the distribution of investment rates determines income classification. This is suitable for categorical data like income group and investment rate level.

### 11.6.3 Results

- Chi-squared statistic ($X^2$): 185.04
- Degrees of freedom: 4
- p-value: ¡ 2.2e-16. This extremely low p-value indicates a highly significant result, well below conventional thresholds (e.g., 0.05 or 0.01).
- Contingency Table Summary:

| | High Income | Low Income | Lower middle income | Upper middle income |
|---|---|---|---|---|
| High Investment | 992 | 156 | 653 | 719 |
| Lower Investment | 800 | 430 | 582 | 739 |

Table 4: Contingency table summary

  High investment rates were most frequent in the High income and Upper middle income groups.

  Low investment rates were notably more frequent in the Low income and Lower middle income groups.

- Expected vs. Observed Differences: The observed distribution of investment categories across income groups differs considerably from what would be expected if the variables were independent.

### 11.6.4 Interpretation

The Chi-squared test results indicate a statistically significant association between a country's income group and its investment rate category. With a p-value far below 0.05, we reject the null hypothesis$H_0$ of independence.

This means the likelihood of a country having high or low investment is not evenly distributed across income groups. In other words, investment behavior appears to be systematically related to income classification, suggesting a dependency between these two categorical variables.

# 12 Conclusions

This project analyzed the relationship between lagged investment rates and GDP per capita growth using a global dataset from 1993 to 2023. The initial hypothesis—whether higher investment leads to higher economic growth—was tested using multiple statistical methods.

First, a simple linear regression showed no significant relationship between the 3-year lagged investment rate and GDP per capita growth, with an R-squared of 0.0029 and a p-value of 0.78. This result led to a failure to reject the null hypothesis of no significant association.

Further analysis using multivariate regressions revealed that the effectiveness of investment varies considerably by income level and region. High-income countries demonstrated the strongest positive returns to investment, while low-income countries often experienced weak or even negative returns, especially when investment rates fell below 15–20% of GDP. Middle-income countries showed intermediate effects.

Regionally, East Asia & Pacific and South Asia exhibited the highest marginal returns to investment, while Sub-Saharan Africa and Latin America showed lower or inconsistent effects, influenced by structural and macroeconomic factors.

Correlation analysis supported these findings: among 180 countries, most showed no significant correlation between investment and growth. Only 5 countries showed strong correlation, while 113 showed none. Across income groups, only low-income countries had a weak correlation; the others had very weak or no correlation.

A two-sample t-test found no statistically significant difference in effective investment rates between high- and low-income countries ($p = 0.7285$), despite slightly higher mean rates in low-income countries. However, a chi-squared test showed a highly significant association between income group and investment rate category, indicating that investment patterns are not randomly distributed across

income levels.

Overall, the results suggest that investment alone does not reliably predict GDP per capita growth. Its effectiveness is context-dependent, shaped by institutional quality, regional dynamics, and income level. Therefore, policy efforts should focus not only on increasing investment but also on improving its efficiency and supporting institutional development, particularly in lower-income and underperforming regions.